Lab experiment

3.Implementation of buffer management

Aim:

To write a C program to implement buffer management by accepting data from the user and displaying the data stored in the buffer, buffer size, and data size.

Explanation:

1.`char buffer[100]` allocates a buffer of 100 bytes.

2.`fgets()` reads the input from the user into the buffer.

3.`strlen()` calculates the length of the input string.

4.`sizeof(buffer)` returns the total buffer capacity, which is 100 bytes.

5.`+1` includes the null terminator (`'\0'`) in the data size.

Coding:

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
   int source, destination;
   char buffer[100];
   ssize_t bytes;

   source = open("source.txt", O_RDONLY);

   if (source < 0)
   {
      perror("Error opening source file");
      return 1;
   }

   destination = open("destination.txt",
                  O_WRONLY | O_CREAT | O_TRUNC,
                  0644);

   if (destination < 0)
   {
      perror("Error opening destination file");
```

```
        close(source);
        return 1;
    }

    printf("File copy program\n");

    while ((bytes = read(source, buffer, sizeof(buffer))) > 0)
    {
        ssize_t total = 0;

        while (total < bytes)
        {
            ssize_t written = write(destination,
                                    buffer + total,
                                    bytes - total);

            if (written < 0)
            {
                perror("Error writing file");
                close(source);
                close(destination);
                return 1;
            }

            total += written;
        }
    }

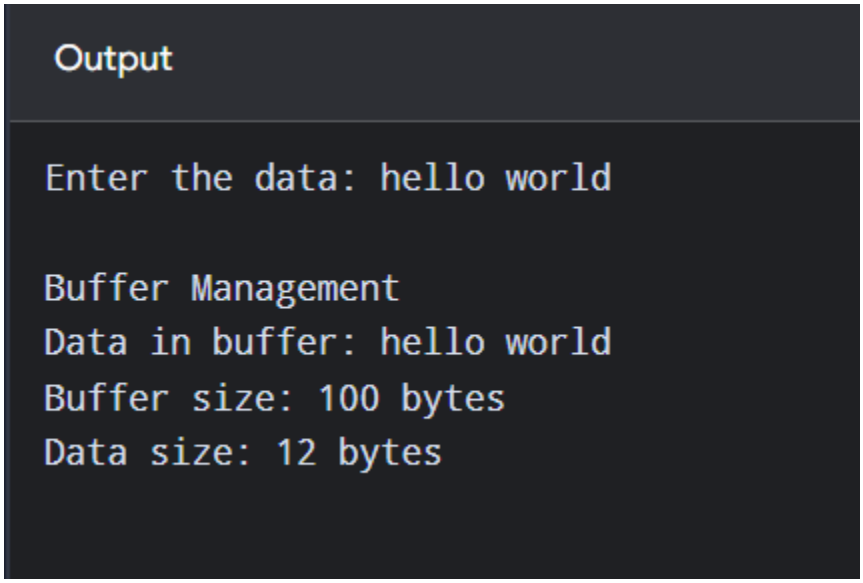
    if (bytes < 0)
    {
        perror("Error reading file");
        close(source);
        close(destination);
        return 1;
    }

    close(source);
    close(destination);

    printf("File copied successfully\n");

    return 0;
}
```

Output:



Result:

Thus, buffer management was implemented successfully in C by storing user input in a buffer and displaying the buffer contents, buffer capacity, and data size.